CELL BROADCAST EMERGENCY ALERT SYSTEM

CMAS / ETWS

Complete Implementation Guide

Open5GS 4G VoLTE + srsRAN + bladeRF xA4

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3GPP TS 23.041 | 3GPP TS 36.331 | ETSI TS 102 900

Tested on: iPhone (eSIM) + Samsung Galaxy (F2G SIM)

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Chapter 1

Introduction

1.1 Document Objective

This document provides a complete and detailed guide to configure a Cell Broadcast emergency alert system (CMAS/ETWS) on a private 4G network using:

- **Open5GS** — EPC Core Network (MME, HSS, SGW, PGW, PCRF)
- **srsRAN 4G (srsENB)** — LTE Base Station with PWS support
- **bladeRF xA4** — SDR for radio transmission
- **Docker** — Containerization of all components

1.2 What is Cell Broadcast?

Cell Broadcast Service (CBS) is a message broadcast service to all mobile terminals within a defined geographic area. Unlike SMS:

Feature	**Cell Broadcast**	**SMS**
Delivery	Instant to ALL	One by one
Network Load	Minimal	High (congestion)
1 Million Phones	$<$ 10 seconds	Hours/Days
During Emergency	Always works	Often fails
Anonymous	Yes	No
Roaming Users	Yes	Limited

Table 1.1: Cell Broadcast vs SMS Comparison

1.3 3GPP Standards

The implemented system complies with the following specifications:

- **3GPP TS 23.041** — Technical realization of Cell Broadcast Service (CBS)
- **3GPP TS 36.331** — E-UTRA RRC Protocol (SIB10, SIB11, SIB12)

- **3GPP TS 23.038** — Alphabets and language-specific information
- **ETSI TS 102 900** — EU-Alert requirements
- **ATIS-0700008** — CMAS (Commercial Mobile Alert System) - US

1.4 Supported Alert Types

1.4.1 ETWS (Earthquake and Tsunami Warning System)

Defined in 3GPP TS 23.041, broadcast via SIB11:

Message ID	Hex	Alert Type
4352	0x1100	Earthquake
4353	0x1101	Tsunami
4354	0x1102	Earthquake + Tsunami
4355	0x1103	ETWS Test

Table 1.2: ETWS Message IDs (SIB11)

1.4.2 CMAS (Commercial Mobile Alert System)

Defined in 3GPP TS 23.041 Annex B, broadcast via SIB12:

Message ID	Hex	Alert Type	Can Disable?
4370	0x1112	Presidential Alert	NO
4371	0x1113	Extreme Threat (Immediate)	Yes
4372	0x1114	Extreme Threat (Expected)	Yes
4373	0x1115	Severe Threat (Immediate)	Yes
4374-4378	0x1116-111A	Severe Threat (Expected)	Yes
4379	0x111B	AMBER Alert	Yes
4380	0x111C	Required Monthly Test	Yes
4381	0x111D	Exercise	Yes
4383-4391	0x111F-1127	Public Safety	Yes

Table 1.3: CMAS Message IDs (SIB12)

Important: The Presidential Alert (Message ID 4370/0x1112) CANNOT be disabled by the user. It is the only alert type guaranteed to reach ALL phones.

Chapter 2

System Architecture

2.1 Overview

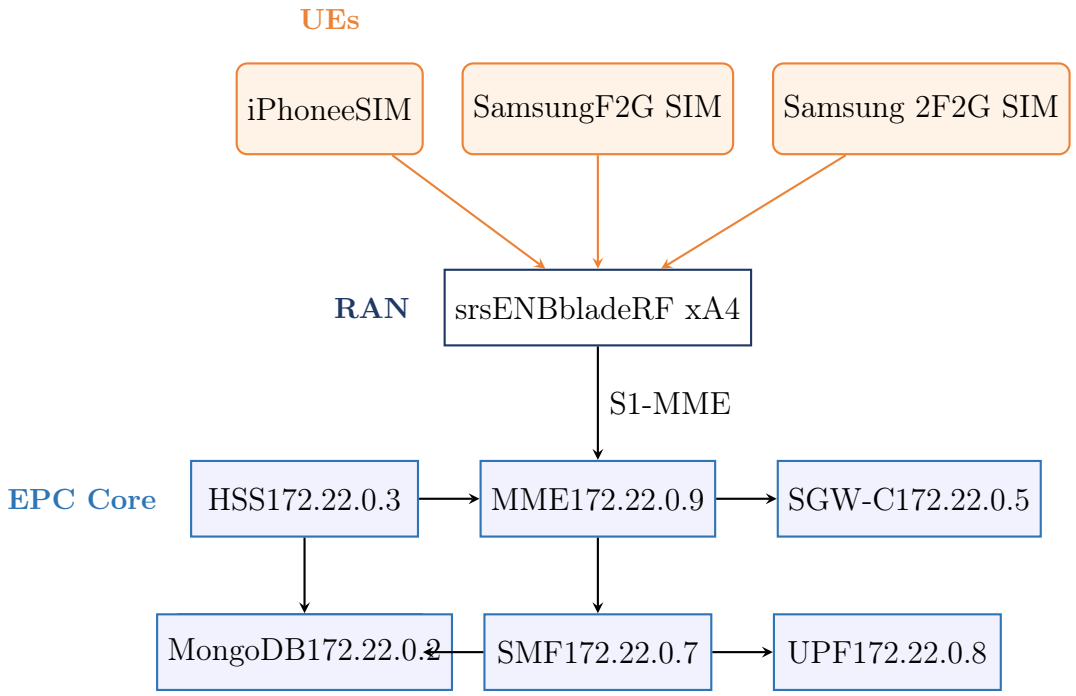


Figure 2.1: Open5GS + srsRAN Architecture

2.2 IP Addressing

Component	IP Address	Port/Protocol
MongoDB	172.22.0.2	27017/tcp
HSS	172.22.0.3	3868/tcp (Diameter)
PCRF	172.22.0.4	3873/tcp
SGW-C	172.22.0.5	2123/udp (GTPv2-C)
SGW-U	172.22.0.6	2152/udp (GTP-U)
SMF	172.22.0.7	8805/udp (PFCP)
UPF	172.22.0.8	8805/udp (PFCP)
MME	**172.22.0.9**	**36412/sctp (S1AP)**
WebUI	172.22.0.26	9999/tcp

Table 2.1: Open5GS Component IP Addresses

2.3 PLMN Parameters

Parameter	Value
MCC (Mobile Country Code)	001
MNC (Mobile Network Code)	01
PLMN ID	00101
TAC (Tracking Area Code)	1
eNB ID	0x19B (411)
Cell ID	Calculated automatically

Table 2.2: PLMN Parameters

Chapter 3

Open5GS Configuration (Core Network)

3.1 Prerequisites

3.1.1 Docker Installation

```
# Install Docker on Ubuntu
sudo apt update
sudo apt install -y docker.io docker-compose
sudo usermod -aG docker $USER
newgrp docker

# Verify
docker --version
docker-compose --version
```

3.1.2 Clone the Repository

```
cd /home/f2g/Telecom/Core-Network/
git clone https://github.com/herlesupreeth/docker_open5gs.git
cd docker_open5gs
```

3.2 Environment Configuration (.env)

File: /home/f2g/Telecom/Core-Network/docker_open5gs/.env

```
# PLMN Configuration
MCC=001
MNC=01
TAC=1

# Network
TEST_NETWORK=172.22.0.0/24
DOCKER_HOST_IP=192.168.1.155

# MongoDB
```

```
MONGO_IP=172.22.0.2

# HSS
HSS_IP=172.22.0.3

# PCRF
PCRF_IP=172.22.0.4
PCRF_BIND_PORT=3873

# SGW
SGWC_IP=172.22.0.5
SGWU_IP=172.22.0.6

# SMF
SMF_IP=172.22.0.7
SMF_DNS1=8.8.8.8
SMF_DNS2=8.8.4.4

# UPF
UPF_IP=172.22.0.8

# MME
MME_IP=172.22.0.9

# UE IP Pools
UE_IPV4_INTERNET=192.168.100.0/24
UE_IPV4_IMS=192.168.101.0/24
MAX_NUM_UE=1024
```

3.3 MME Configuration

File: /home/f2g/Telecom/Core-Network/docker_open5gs/mme/mme.yaml

```
logger:
    file:
      path: /open5gs/install/var/log/open5gs/mme.log

global:
  max:
    ue: MAX_NUM_UE

mme:
    freeDiameter: /open5gs/install/etc/freeDiameter/mme.conf
    s1ap:
      server:
        - dev: MME_IF
    gtpc:
      server:
        - dev: MME_IF
      client:
```

```
        sgwc:
          - address: SGWC_IP
        smf:
          - address: SMF_IP
    gummei:
      - plmn_id:
          mcc: MCC
          mnc: MNC
        mme_gid: 2
        mme_code: 1
    tai:
      - plmn_id:
          mcc: MCC
          mnc: MNC
        tac: TAC
    security:
        integrity_order : [ EIA2, EIA1, EIA0 ]
        ciphering_order : [ EEA0, EEA1, EEA2 ]
    network_name:
        full: K48 Telecommunications
        short: K48
    mme_name: F2G-mme0
```

3.4 Starting Open5GS

```
cd /home/f2g/Telecom/Core-Network/docker_open5gs

# Build the Docker image
docker-compose -f 4g-volte-deploy.yaml build

# Start all containers
docker-compose -f 4g-volte-deploy.yaml up -d

# Check status
docker ps --format "table {{.Names}}\t{{.Status}}\t{{.Ports}}"

# View MME logs (important for seeing attachments)
docker logs -f mme
```

3.5 HSS Configuration (Subscribers)

3.5.1 Adding F2G SIMs to MongoDB

```
# Connect to MongoDB
docker exec -it mongo mongosh open5gs

# Add a subscriber (example SIM 1)
```

```
db.subscribers.insertOne({
  "imsi": "001010000123450",
  "msisdn": ["1003"],
  "security": {
    "k": "B4B066FA5A5DD3C1E66FE0A1EBD81A62",
    "op": null,
    "opc": "772C8F93FC0FF3F39DD2DF9AA70B5CE6",
    "amf": "8000"
  },
  "ambr": {
    "downlink": {"value": 1, "unit": 3},
    "uplink": {"value": 1, "unit": 3}
  },
  "slice": [{
    "sst": 1,
    "default_indicator": true,
    "session": [{
      "name": "internet",
      "type": 3,
      "qos": {"index": 9, "arp": {"priority_level": 8}},
      "ambr": {
        "downlink": {"value": 1, "unit": 3},
        "uplink": {"value": 1, "unit": 3}
      }
    }]
  }],
  "access_restriction_data": 32,
  "subscriber_status": 0,
  "network_access_mode": 0
})
```

3.5.2 Configured SIMs List

IMSI	MSISDN	Type	Ki
001010000123450	1003	F2G SIM	B4B066FA5A5DD3C1...
001010000123451	1001	F2G SIM	B4B066FA5A5DD3C1...
001010000123452	1002	F2G SIM	B4B066FA5A5DD3C1...
001010000123453	1004	F2G SIM	B4B066FA5A5DD3C1...
001010000099901	1099	eSIM Simlessly	B4B066FA5A5DD3C1...

Table 3.1: Subscribers Configured in HSS

Chapter 4

srsRAN Configuration (eNB)

4.1 Prerequisites

4.1.1 srsRAN 4G Installation

```
# Dependencies
sudo apt install -y cmake build-essential libfftw3-dev \
    libmbedtls-dev libsctp-dev libyaml-cpp-dev libgtest-dev \
    libboost-all-dev libconfig++-dev

# Clone and compile srsRAN 4G
cd /home/f2g/build
git clone https://github.com/srsran/srsRAN_4G.git
cd srsRAN_4G
mkdir build && cd build
cmake ../ -DCMAKE_BUILD_TYPE=Release
make -j$(nproc)
sudo make install
sudo ldconfig

# Install default config files
sudo srsran_install_configs.sh
```

4.1.2 bladeRF xA4 Configuration

```
# Install bladeRF
sudo apt install -y bladerf libbladerf-dev bladerf-fpga-hostedx40

# Load FPGA
sudo bladeRF-cli -l /usr/share/Nuand/bladeRF/hostedxA4.rbf

# Verify
bladeRF-cli -p
# Should display: Backend: libusb, Serial: ...
```

4.2 eNB Configuration with PWS Support

File: /home/f2g/Telecom/Core-Network/docker_open5gs/srsran/enb_pws.conf

```
##############################################################################
# srsENB Configuration with PWS (CMAS/ETWS) Support
# F2G TelcoLab - PKF Telco Academy
##############################################################################

[enb]
enb_id = 0x19B
mcc = 001
mnc = 01
mme_addr = 172.22.0.9
gtp_bind_addr = 172.22.0.1
s1c_bind_addr = 172.22.0.1
n_prb = 50
nof_ports = 1

[enb_files]
sib_config =
   /home/f2g/Telecom/Core-Network/docker_open5gs/srsran/sib_complete_pws.conf
rr_config = /etc/srsran/rr.conf
rb_config = /etc/srsran/rb.conf

[rf]
dl_earfcn = 1450
tx_gain = 60
rx_gain = 40
device_name = bladeRF

[log]
all_level = info
filename = /tmp/enb_pws.log

[expert]
pusch_max_its = 8
nof_phy_threads = 3
tx_amplitude = 0.6
```

4.2.1 Radio Parameters Explained

Parameter	Value	Meaning
dl_earfcn	1450	EARFCN Band 3 (1800 MHz)
		DL: 1830 MHz, UL: 1735 MHz
n_prb	50	50 PRB = 10 MHz bandwidth
tx_gain	60	Transmission gain (dB)
rx_gain	40	Reception gain (dB)
tx_amplitude	0.6	TX amplitude (0-1)

Table 4.1: srsENB Radio Parameters

4.3 SIB Configuration for PWS

File: `/home/f2g/Telecom/Core-Network/docker_open5gs/srsran/sib_complete_pws.conf`

4.3.1 SIB1 - System Information Block 1

SIB1 defines the scheduling of System Information, including SIB11 (ETWS) and SIB12 (CMAS):

```
sib1 =
{
    intra_freq_reselection = "Allowed";
    q_rx_lev_min = -65;
    cell_barred = "NotBarred"
    si_window_length = 20;
    sched_info =
    (
        {
            # SI-1: SIB3 (cell reselection info)
            si_periodicity = 16;
            si_mapping_info = [ 3 ];
        },
        {
            # SI-2: SIB11 (ETWS) - Emergency alerts
            si_periodicity = 64;
            si_mapping_info = [ 11 ];
        },
        {
            # SI-3: SIB12 (CMAS) - Commercial mobile alerts
            si_periodicity = 64;
            si_mapping_info = [ 12 ];
        }
    );
    system_info_value_tag = 0;
};
```

4.3.2 SIB11 - ETWS (Earthquake and Tsunami Warning)

```
########################################################################
# SIB11 - ETWS (Earthquake and Tsunami Warning System)
# 3GPP TS 36.331, TS 23.041
#
# Message IDs:
#   0x1100 (4352) = Earthquake
#   0x1101 (4353) = Tsunami
#   0x1102 (4354) = Earthquake + Tsunami
#   0x1103 (4355) = Test
########################################################################
sib11 =
{
    # Message Identifier - 0x1100 = Earthquake
    message_id = 0x1100;

    # Serial Number (unique per message)
    serial_num = 0x3210;

    # Data Coding Scheme (GSM-7)
    data_coding_scheme = 0x01;

    # Warning message (max ~90 chars GSM-7)
    warning_message = "EARTHQUAKE ALERT - Earthquake detected.
                       Take cover immediately.";
};
```

4.3.3 SIB12 - CMAS (Commercial Mobile Alert System)

```
########################################################################
# SIB12 - CMAS (Commercial Mobile Alert System)
# 3GPP TS 36.331, TS 23.041
#
# Message IDs (3GPP TS 23.041 Annex B):
#   4370 (0x1112) = Presidential Alert (CANNOT be disabled)
#   4379 (0x111B) = AMBER Alert
#   4380 (0x111C) = Required Monthly Test
########################################################################
sib12 =
{
    # Message Identifier - 0x1112 = Presidential Alert
    message_id = 0x1112;

    # Serial Number (unique per message)
    serial_num = 0x32A7;

    # Data Coding Scheme (GSM-7)
    data_coding_scheme = 0x01;

```

```
    # Warning message
    warning_message = "NATIONAL ALERT - This is an emergency
                       broadcast. All phones receive this alert!";
};
```

4.4 Starting srsENB

```
# Verify bladeRF is connected
bladeRF-cli -p

# Start srsENB with PWS config
cd /home/f2g/Telecom/Core-Network/docker_open5gs
sudo srsenb srsran/enb_pws.conf

# Or in background with logs
nohup sudo srsenb srsran/enb_pws.conf > /tmp/srsenb_pws.log 2>&1 &

# Check logs
tail -f /tmp/srsenb_pws.log
```

4.4.1 Verifying Operation

Expected log messages:

```
---  Software Radio Systems LTE eNodeB  ---

Reading configuration file srsran/enb_pws.conf...
Opening bladeRF...
Set RX gain to 40.0 dB
Set TX gain to 60.0 dB

==== eNodeB started ===
Type <t> to view trace

Connecting to MME 172.22.0.9:36412
SCTP Association established
S1 Setup Response received
```

Chapter 5

Sending Alerts

5.1 Alert Sending Script

The `send_pws_alert.sh` script allows sending CMAS/ETWS alerts by modifying the SIB configuration and restarting srsENB.

File: `/home/f2g/Telecom/Core-Network/docker_open5gs/scripts/send_pws_alert.sh`

5.1.1 Usage

```
# Syntax
./scripts/send_pws_alert.sh <type> "<message>"

# Available alert types:
#   CMAS (SIB12):
#     test          - Required Monthly Test (4380)
#     presidential  - Presidential Alert (4370) - Cannot be disabled
#     extreme       - Extreme Threat (4371)
#     severe        - Severe Threat (4373)
#     amber         - AMBER Alert - Child abduction (4379)
#
#   ETWS (SIB11):
#     earthquake    - Earthquake Warning (4352)
#     tsunami       - Tsunami Warning (4353)
```

5.2 Alert Sending Examples

5.2.1 Presidential Alert (cannot be disabled)

```
./scripts/send_pws_alert.sh presidential \
    "NATIONAL ALERT - This is an urgent government message."

# Output:
# ====================================================
#   SENDING ALERT: PRESIDENTIAL ALERT
# ====================================================
```

```
# Type: PRESIDENTIAL ALERT
# SIB: SIB12
# Message ID: 0x1112
# ...
# ALERT BROADCAST VIA SIB12
```

5.2.2 Earthquake Alert (ETWS)

```
./scripts/send_pws_alert.sh earthquake \
    "EARTHQUAKE - Magnitude 6.2 detected. Evacuate immediately."

# This alert uses SIB11 (ETWS)
# Message ID: 0x1100
```

5.2.3 Tsunami Alert (ETWS)

```
./scripts/send_pws_alert.sh tsunami \
    "TSUNAMI ALERT - Move to higher ground immediately."
```

5.2.4 AMBER Alert (Child Abduction)

```
./scripts/send_pws_alert.sh amber \
    "AMBER ALERT - Missing child. Vehicle: White Toyota AB-123-CD"
```

5.2.5 Monthly Test

```
./scripts/send_pws_alert.sh test \
    "TEST - This is a test of the emergency alert system."
```

5.3 Technical Operation

The alert sending process follows these steps:

1. **SIB Modification**: The script modifies the `sib_complete_pws.conf` file to update:
   - The `message_id` corresponding to the alert type
   - The `serial_num` (unique for each send)
   - The `warning_message` with the alert text
2. **srsENB Restart**: srsENB must be restarted to load the new SIB configuration
3. **Broadcast**: srsENB automatically broadcasts SIB11 or SIB12 according to configured periodicity (64 radio frames = 640 ms)
4. **Reception**: UEs in the cell receive the SIB and display the alert

Chapter 6

Troubleshooting

6.1 Common Issues

6.1.1 srsENB Doesn't Connect to MME

Symptom: "S1 Setup Request failed" or timeout

Solutions:

```
# Check that MME is running
docker ps | grep mme
docker logs mme

# Check connectivity
ping 172.22.0.9

# Check SCTP port is open
sudo ss -tlnp | grep 36412

# Restart MME if necessary
docker restart mme
```

6.1.2 UE Doesn't Attach

Symptom: "Authentication failure" or "Attach reject"

Solutions:

```
# Check subscriber exists in MongoDB
docker exec -it mongo mongosh open5gs
db.subscribers.find({imsi: "001010000123450"})

# Check HSS logs
docker logs hss | grep -i auth

# Verify Ki/OPc in MongoDB vs SIM
```

6.1.3 Alerts Not Received on Phone

Symptom: srsENB broadcasts but phone doesn't display anything

Solutions:

1. Verify phone is attached (check MME logs)
2. Use Presidential Alert (0x1112) which cannot be disabled
3. On Samsung: check Settings → Safety and emergency → Wireless emergency alerts
4. Verify SIB11/12 is loaded in srsENB logs

6.1.4 bladeRF Not Detected

```
# Check USB connection
lsusb | grep Nuand
# Expected: Bus xxx Device xxx: ID 2cf0:5250 Nuand LLC bladeRF 2.0

# Load FPGA
sudo bladeRF-cli -l /usr/share/Nuand/bladeRF/hostedxA4.rbf

# Check permissions
sudo chmod 666 /dev/bus/usb/00X/00Y
```

6.2 Useful Logs

```
# srsENB logs
tail -f /tmp/enb_pws.log

# MME logs (attachments)
docker logs -f mme | grep -E "(Attach|UE)"

# HSS logs (authentication)
docker logs -f hss

# Sent alerts log
cat /tmp/pws_alerts.log
```

Chapter 7

References

7.1 3GPP Specifications

- **3GPP TS 23.041** — Technical realization of Cell Broadcast Service (CBS)
- **3GPP TS 36.331** — E-UTRA Radio Resource Control (RRC); Protocol specification
- **3GPP TS 23.038** — Alphabets and language-specific information
- **3GPP TS 36.304** — E-UTRA; UE procedures in idle mode

7.2 Regional Standards

- **ATIS-0700008** — CMAS Commercial Mobile Alert System (US)
- **ETSI TS 102 900** — EU-Alert system requirements
- **J-ALERT** — Japan Alert System (ETWS)

7.3 Software Resources

- Open5GS: https://open5gs.org/
- srsRAN: https://www.srsran.com/
- bladeRF: https://www.nuand.com/
- Docker Open5GS: https://github.com/herlesupreeth/docker_open5gs

Appendix A

Appendices

A.1 Complete Message ID Table

ID	Hex	Description	SIB
4352	0x1100	ETWS Earthquake	SIB11
4353	0x1101	ETWS Tsunami	SIB11
4354	0x1102	ETWS Earthquake + Tsunami	SIB11
4355	0x1103	ETWS Test	SIB11
4370	0x1112	Presidential Alert	SIB12
4371	0x1113	Extreme Threat (Immediate)	SIB12
4372	0x1114	Extreme Threat (Expected)	SIB12
4373	0x1115	Severe Threat (Immediate)	SIB12
4374	0x1116	Severe Threat (Expected)	SIB12
4379	0x111B	AMBER Alert	SIB12
4380	0x111C	Required Monthly Test	SIB12
4381	0x111D	Exercise	SIB12
4383-4391	0x111F-1127	Public Safety	SIB12

Table A.1: Complete CMAS/ETWS Message IDs

A.2 Deployment Checklist

Docker and Docker Compose installed

docker_open5gs repository cloned

.env file configured with correct IPs

Open5GS Docker containers started (19 containers)

Subscribers added to MongoDB (HSS)

srsRAN 4G compiled and installed

bladeRF connected and FPGA loaded

enb_pws.conf configured

sib_complete_pws.conf configured

srsENB connected to MME

Phone attached to network

Test alert sent and received

Document prepared by Arnaud DJOUM — F2G TelcoLab
For PKF Telco Academy Cameroon
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